Territory Reconfiguration Web App - GCP VM Deployment

Native Python Server (server.py), Systemd & NGINX via PuTTY (Zero Streamlit / No Docker)

Target: GCP Compute Engine (Ubuntu LTS)
SSH Client: PuTTY & PuTTYgen | v.0.8.4

1. WINDOWS ADMIN	->	2. CUSTOM DOMAIN	->	3. NGINX PROXY	->	4. NATIVE SERVER.PY	->	5. ERPNEXT V15
PuTTY SSH (Port 22)		DNS A-Record -> VM IP		Ports 80 / 443 (SSL)		127.0.0.1:8765 (24/7)		REST API Backend

PHASE 1 GCP Console VM & Network Configuration

GCP WEB CONSOLE

1. **Compute Engine VM:** Create or select an Ubuntu 22.04 / 24.04 LTS VM (e2-medium, 2 vCPU / 4 GB RAM, asia-southeast1).
2. **Firewall Rules:** Under VM Firewall settings, enable both Allow HTTP traffic (Port 80) and Allow HTTPS traffic (Port 443).
3. **Static External IP:** Go to VPC network -> IP addresses, find the VM external IP, and click Promote to static IP.

PHASE 2 SSH Key Generation (PuTTYgen) & VM Connection (PuTTY)

WINDOWS WORKSTATION

STEP A: GENERATE KEY IN PUTTYGEN

1. **Key Type:** Select RSA (2048-bit) and click Generate.
2. **Key Comment:** Set comment to Linux user (e.g., pimsadmin).
3. **Private Key:** Click Save private key as gcp-territory-vm.ppk.
4. **Public Key:** Copy the top ssh-rsa AAAA... text block.

STEP B: REGISTER IN GCP & CONNECT IN PUTTY

1. **GCP VM Edit:** VM instances -> Edit -> SSH Keys -> Add Item.
2. **Save Key:** Paste public key text block and click Save.
3. **PuTTY Host:** Set Host to pimsadmin@YOUR_VM_IP (Port 22).
4. **Credentials:** SSH -> Auth -> Credentials -> Select .ppk file.

PHASE 3 Native Application Setup on VM (Zero Streamlit Dependency)

PUTTY SSH TERMINAL

1. INSTALL SYSTEM PACKAGES (PYTHON 3 STANDARD LIB, GIT, NGINX, CERTBOT)

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y python3 git nginx certbot python3-certbot-nginx
```

2. DEPLOY NATIVE WEB APP FILES (server.py, secure_frappe_gateway.py, territory_reconfiguration_portal.html)

```
sudo mkdir -p /var/www/territory-portal && sudo chown -R $USER:$USER /var/www/territory-portal
git clone https://github.com/LetscodeJosh/HCP_Test_Repo.git /tmp/hcp-repo
cp /tmp/hcp-repo/installers/Territory_Reconfiguration_App/*.py /var/www/territory-portal/
cp /tmp/hcp-repo/installers/Territory_Reconfiguration_App/*.html /var/www/territory-portal/
rm -rf /tmp/hcp-repo
```

3. CONFIGURE STANDARD LINUX ENVIRONMENT FILE (/var/www/territory-portal/.env)

```
cat << 'EOF' > /var/www/territory-portal/.env
ERPNEXT_URL=https://dev.pmii-marketing.com
EOF
```

4. OPTIONAL SANITY CHECK (VERIFY NATIVE SERVER.PY ON PORT 8765)

```
cd /var/www/territory-portal && python3 server.py
# Listens on http://127.0.0.1:8765 (Press Ctrl+C after test, then proceed to Phase 4 Systemd Service)
```

Background Service, Domain Proxy & SSL Activation

24/7 Systemd Daemon (Port 8765), NGINX Reverse Proxy, Let's Encrypt HTTPS & Day-2 Operations

Internal Service: 127.0.0.1:8765

Public Ports: 80 (HTTP) / 443 (HTTPS)

PHASE 4 — 24/7 Background Daemon (Systemd Service)

KEEPS APP RUNNING AFTER CLOSING PUTTY

1. CREATE SYSTEMD UNIT: /etc/systemd/system/territory-portal.service

```
sudo tee /etc/systemd/system/territory-portal.service > /dev/null << 'EOF'
[Unit]
Description=PIMS Territory Reconfiguration Native Web Server
After=network.target

[Service]
Type=simple
User=root
WorkingDirectory=/var/www/territory-portal
EnvironmentFile=/var/www/territory-portal/.env
ExecStart=/usr/bin/python3 /var/www/territory-portal/server.py
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
EOF
```

2. ENABLE & START BACKGROUND SERVICE

```
sudo systemctl daemon-reload && sudo systemctl enable territory-portal && sudo systemctl start territory-portal
```

PHASE 5 — Custom Domain DNS, NGINX Reverse Proxy & HTTPS SSL

DOMAIN & SSL SETUP

1. **DNS A-Record:** In your domain DNS provider, point your subdomain (e.g., territory.yourdomain.com) to YOUR_VM_STATIC_IP.

2. CREATE NGINX REVERSE PROXY CONFIG: /etc/nginx/sites-available/territory-portal

```
sudo tee /etc/nginx/sites-available/territory-portal > /dev/null << 'EOF'
server {
    listen 80;
    server_name territory.yourdomain.com;
    client_max_body_size 50M;

    location / {
        proxy_pass http://127.0.0.1:8765/;
        proxy_http_version 1.1;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Host $host;
        proxy_read_timeout 300;
    }
}
EOF
```

3. ACTIVATE SITE & PROVISION FREE LET'S ENCRYPT HTTPS CERTIFICATE

```
sudo ln -sf /etc/nginx/sites-available/territory-portal /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default && sudo nginx -t && sudo systemctl restart nginx
sudo certbot --nginx -d territory.yourdomain.com
```

PHASE 6 — Developer Operations & Quick Update Reference

DAY-2 MAINTENANCE

OPERATION	PUTTY TERMINAL COMMAND
Restart Web Server	`sudo systemctl restart territory-portal`
Check Service Health	`sudo systemctl status territory-portal`
Stream Live App Logs	`sudo journalctl -u territory-portal -f -n 100`
Reload NGINX Proxy	`sudo nginx -t && sudo systemctl restart nginx`